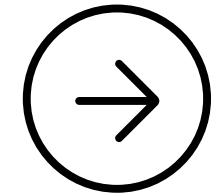


CYBERSECURITY FRAMEWORK FOR **SMALL BUSINESSES**

PRESENTED BY :

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AGENDA



1 Introduction

2 Threat Analysis

3 Framework Selection

4 Network Setup

5 Firewall Configuration

6 Attack Simulation

7 Attack Prevention

8 IDS/IPS (Snort / Suricata)

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10 Conclusion



INTRODUCTION

This project focuses on building a practical cybersecurity framework for small businesses that lack strong security systems. It aims to provide cost-effective protection using structured methods and open-source tools.



Threat Analysis

```
root@kali: /home/kali

(root@kali)-[/home/kali]
# nmap 192.168.1.47
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-31 13:07 EDT
Nmap scan report for 192.168.1.47
Host is up (0.0043s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: 08:00:27:A8:3E:B1 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds

(root@kali)-[/home/kali]
# nmap 192.168.1.72
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-31 13:08 EDT
Nmap scan report for 192.168.1.72
Host is up (0.0023s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: 08:00:27:A8:3E:B1 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

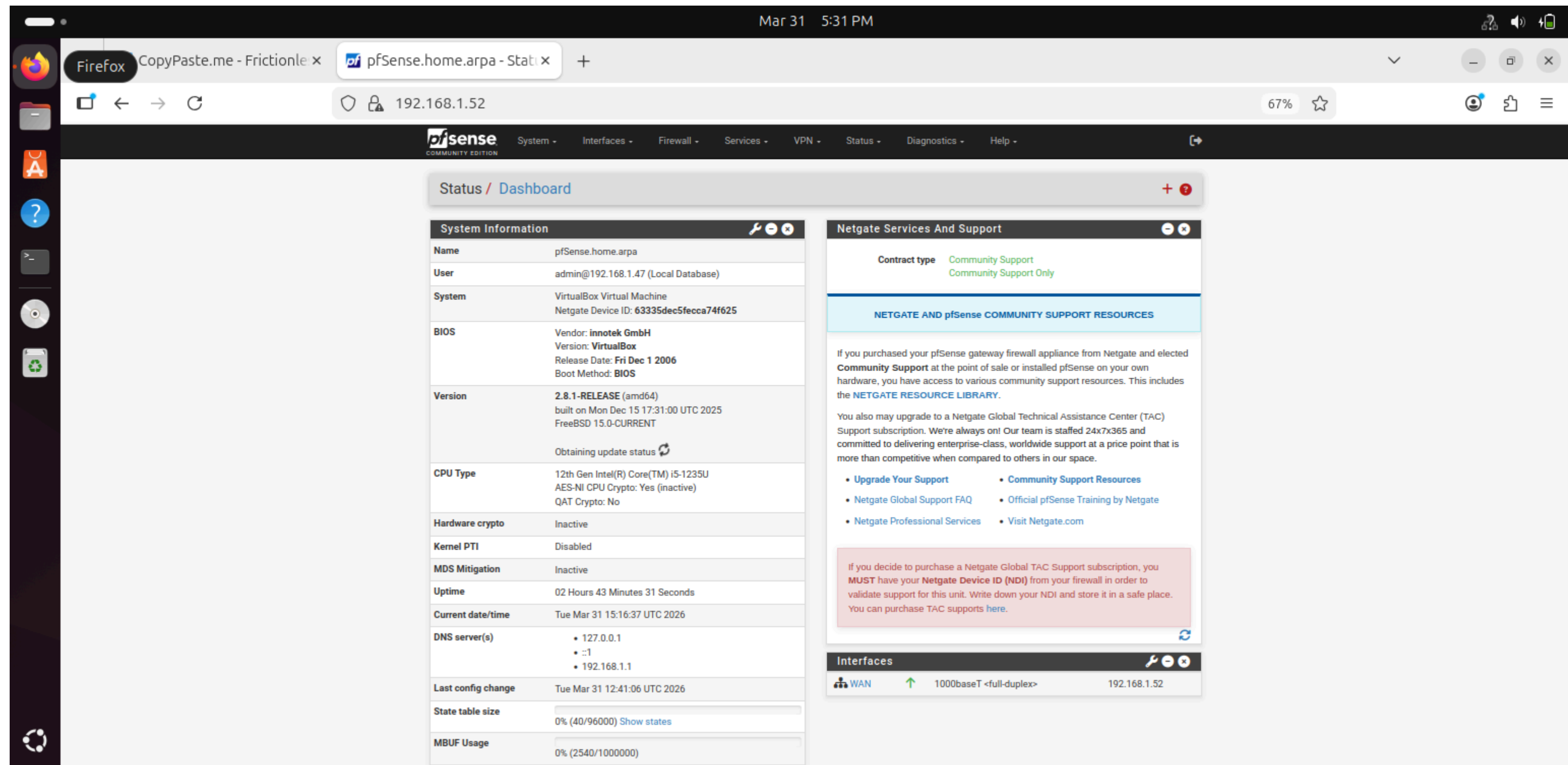
Nmap done: 1 IP address (1 host up) scanned in 0.53 seconds

(root@kali)-[/home/kali]
#
```

***Common threats include unauthorized access, open ports, and network scanning attacks.
Using Nmap, the target system was scanned to identify exposed
services and vulnerabilities.***



Framework Selection



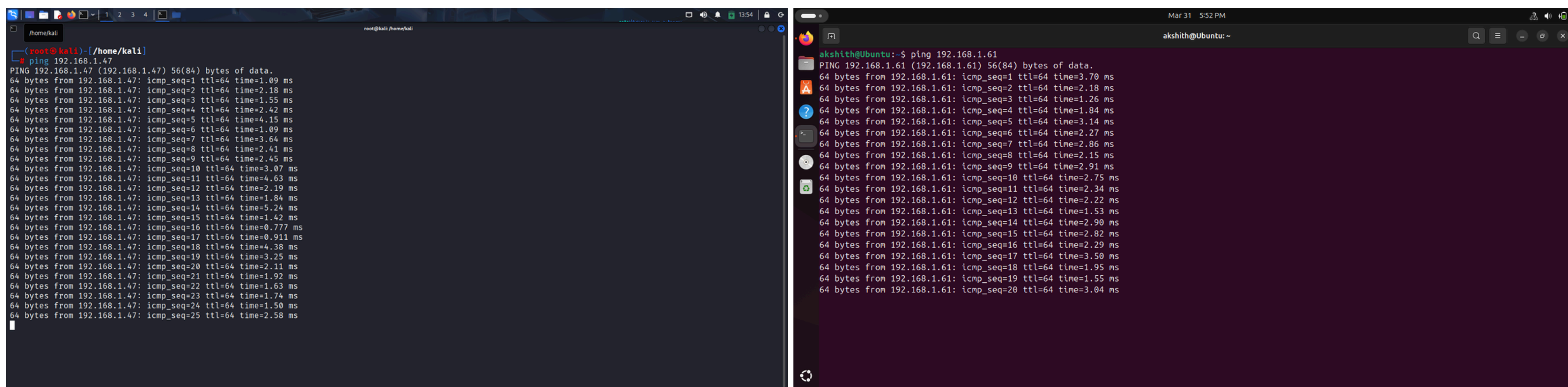
CIS Controls framework is selected as it provides essential security practices for small businesses. It focuses on basic protections like access control, monitoring, and vulnerability management.

Network Setup

Kali IP Address is 192.168.1.61

Ubuntu IP addresses is 192.168.1.47

pfSense as firewall, Kali Linux as attacker, Ubuntu as target system



The image displays two terminal windows side-by-side. The left window is a Kali Linux terminal with a dark blue background, showing a root user at the /home/kali directory. It has executed the command 'ping 192.168.1.47', resulting in 25 successful ping responses from 192.168.1.47 to 192.168.1.61. The right window is an Ubuntu terminal with a dark purple background, showing an akshith user at the ~ directory. It has executed the command 'ping 192.168.1.61', resulting in 20 successful ping responses from 192.168.1.61 to 192.168.1.61. Both windows show detailed ICMP echo request statistics including sequence numbers, TTL, and response times.

```
(root@kali)~[/home/kali]
# ping 192.168.1.47
PING 192.168.1.47 (192.168.1.47) 56(84) bytes of data.
64 bytes from 192.168.1.47: icmp_seq=1 ttl=64 time=1.09 ms
64 bytes from 192.168.1.47: icmp_seq=2 ttl=64 time=2.18 ms
64 bytes from 192.168.1.47: icmp_seq=3 ttl=64 time=1.55 ms
64 bytes from 192.168.1.47: icmp_seq=4 ttl=64 time=2.42 ms
64 bytes from 192.168.1.47: icmp_seq=5 ttl=64 time=4.15 ms
64 bytes from 192.168.1.47: icmp_seq=6 ttl=64 time=1.09 ms
64 bytes from 192.168.1.47: icmp_seq=7 ttl=64 time=3.64 ms
64 bytes from 192.168.1.47: icmp_seq=8 ttl=64 time=2.41 ms
64 bytes from 192.168.1.47: icmp_seq=9 ttl=64 time=2.45 ms
64 bytes from 192.168.1.47: icmp_seq=10 ttl=64 time=3.07 ms
64 bytes from 192.168.1.47: icmp_seq=11 ttl=64 time=4.63 ms
64 bytes from 192.168.1.47: icmp_seq=12 ttl=64 time=2.19 ms
64 bytes from 192.168.1.47: icmp_seq=13 ttl=64 time=1.84 ms
64 bytes from 192.168.1.47: icmp_seq=14 ttl=64 time=5.24 ms
64 bytes from 192.168.1.47: icmp_seq=15 ttl=64 time=1.42 ms
64 bytes from 192.168.1.47: icmp_seq=16 ttl=64 time=0.777 ms
64 bytes from 192.168.1.47: icmp_seq=17 ttl=64 time=0.911 ms
64 bytes from 192.168.1.47: icmp_seq=18 ttl=64 time=4.38 ms
64 bytes from 192.168.1.47: icmp_seq=19 ttl=64 time=3.25 ms
64 bytes from 192.168.1.47: icmp_seq=20 ttl=64 time=2.11 ms
64 bytes from 192.168.1.47: icmp_seq=21 ttl=64 time=1.92 ms
64 bytes from 192.168.1.47: icmp_seq=22 ttl=64 time=1.63 ms
64 bytes from 192.168.1.47: icmp_seq=23 ttl=64 time=1.74 ms
64 bytes from 192.168.1.47: icmp_seq=24 ttl=64 time=1.50 ms
64 bytes from 192.168.1.47: icmp_seq=25 ttl=64 time=2.58 ms
^C
```

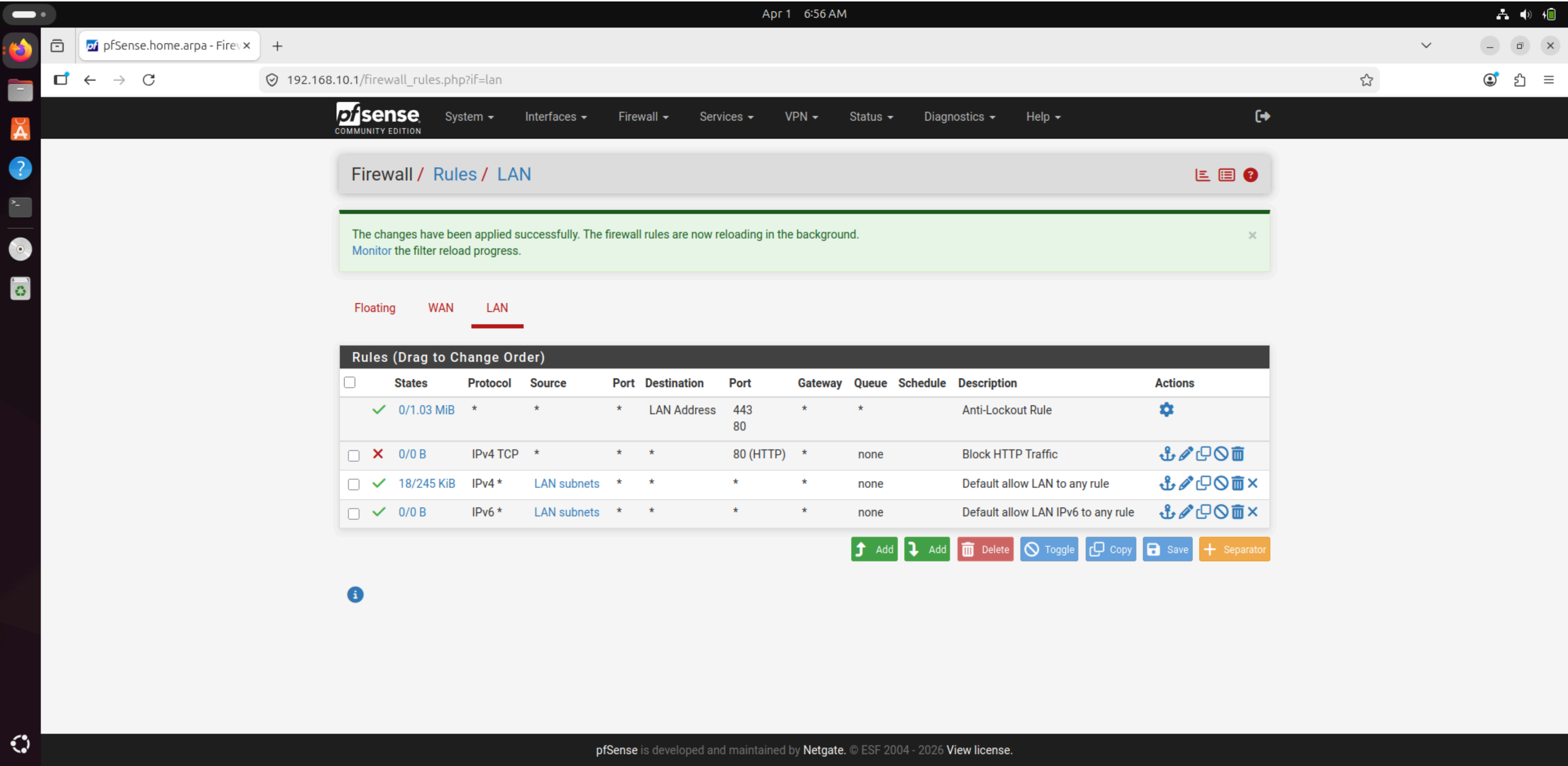
```
akshith@Ubuntu:~$ ping 192.168.1.61
PING 192.168.1.61 (192.168.1.61) 56(84) bytes of data.
64 bytes from 192.168.1.61: icmp_seq=1 ttl=64 time=3.70 ms
64 bytes from 192.168.1.61: icmp_seq=2 ttl=64 time=2.18 ms
64 bytes from 192.168.1.61: icmp_seq=3 ttl=64 time=1.26 ms
64 bytes from 192.168.1.61: icmp_seq=4 ttl=64 time=1.84 ms
64 bytes from 192.168.1.61: icmp_seq=5 ttl=64 time=3.14 ms
64 bytes from 192.168.1.61: icmp_seq=6 ttl=64 time=2.27 ms
64 bytes from 192.168.1.61: icmp_seq=7 ttl=64 time=2.86 ms
64 bytes from 192.168.1.61: icmp_seq=8 ttl=64 time=2.15 ms
64 bytes from 192.168.1.61: icmp_seq=9 ttl=64 time=2.91 ms
64 bytes from 192.168.1.61: icmp_seq=10 ttl=64 time=2.75 ms
64 bytes from 192.168.1.61: icmp_seq=11 ttl=64 time=2.34 ms
64 bytes from 192.168.1.61: icmp_seq=12 ttl=64 time=2.22 ms
64 bytes from 192.168.1.61: icmp_seq=13 ttl=64 time=1.53 ms
64 bytes from 192.168.1.61: icmp_seq=14 ttl=64 time=2.90 ms
64 bytes from 192.168.1.61: icmp_seq=15 ttl=64 time=2.82 ms
64 bytes from 192.168.1.61: icmp_seq=16 ttl=64 time=2.29 ms
64 bytes from 192.168.1.61: icmp_seq=17 ttl=64 time=3.50 ms
64 bytes from 192.168.1.61: icmp_seq=18 ttl=64 time=1.95 ms
64 bytes from 192.168.1.61: icmp_seq=19 ttl=64 time=1.55 ms
64 bytes from 192.168.1.61: icmp_seq=20 ttl=64 time=3.04 ms
```

A virtual network was configured using pfSense as firewall, Kali Linux as attacker, and Ubuntu as target system.

All systems were assigned IP addresses and successfully connected within the same network.

Connectivity was verified using ping between systems.

Firewall Configuration



Firewall rules were configured in pfSense to control network traffic. Specific IP addresses and services were filtered to prevent unauthorized access. This improves network security by restricting malicious connections.

pfSense.home.arpa - Fire x

192.168.10.1/firewall_rules_edit.php?if=lan

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Any

Source Address

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

☐ Invert match

Any

Destination Address

/

Destination Port Range

HTTP (80)

From

Custom

HTTP (80)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

Block HTTP Traffic

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Save

pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.

pfSense.home.arpa - Fire x

192.168.10.1/firewall_rules_edit.php?if=lan

pfSense
COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / Edit

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Any

Source Address

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

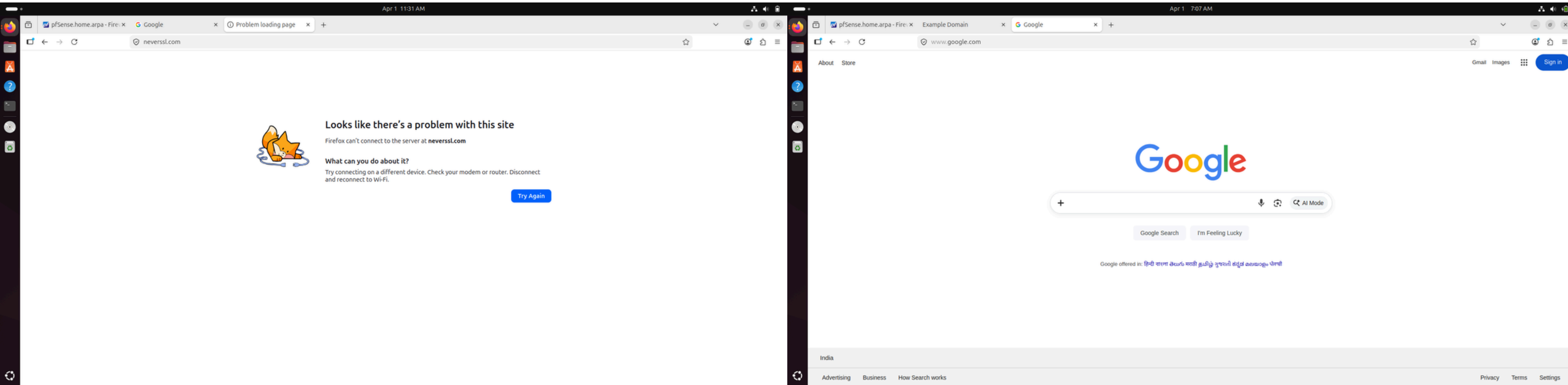
☐ Invert match

Any

Destination Address

/

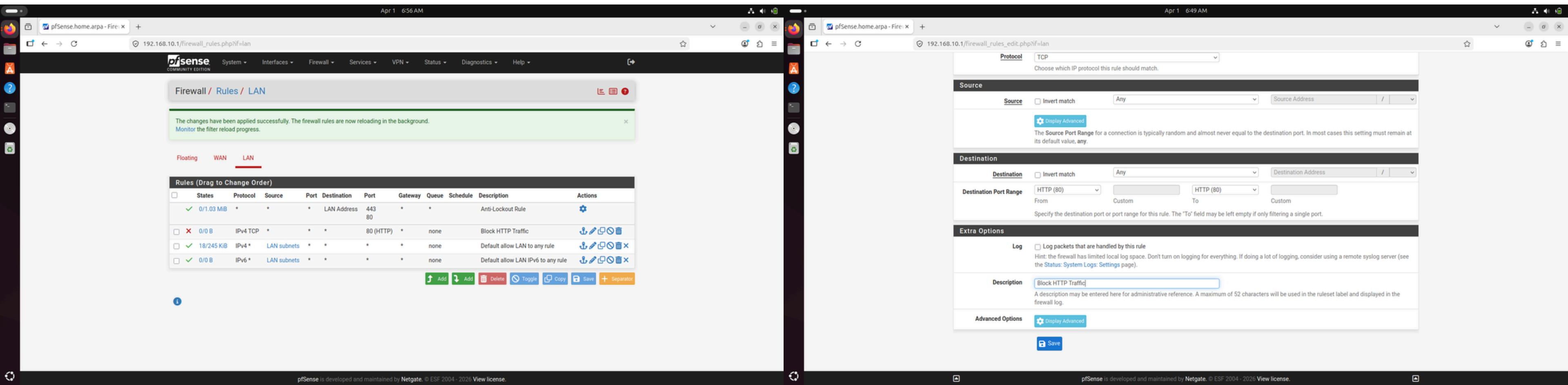
Attack Simulation



http://neverssl.com fails → because the rule blocks port 80 (HTTP
https://google.com works → because it uses port 443 (HTTPS), which did NOT block

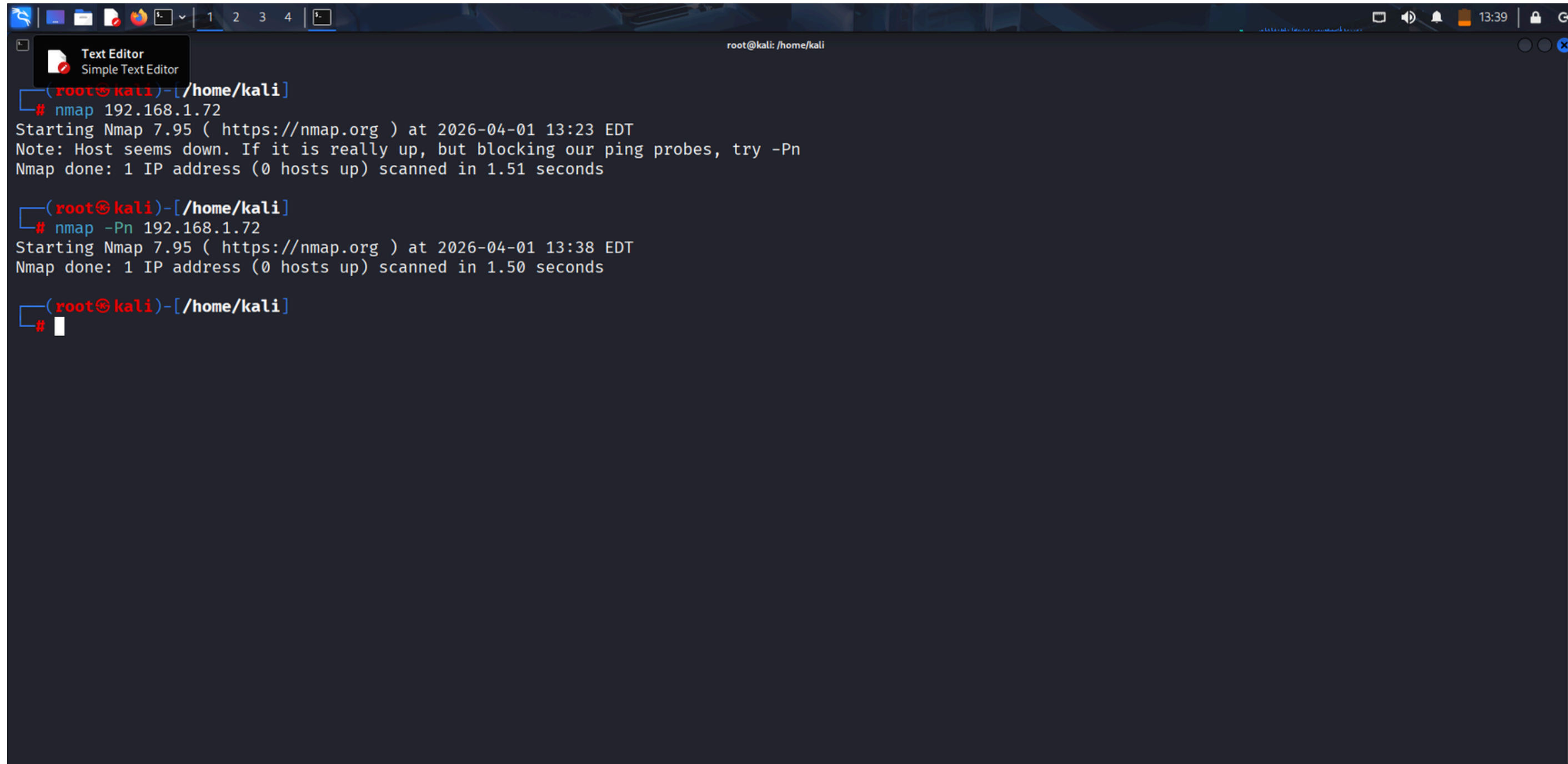


Attack Prevention



Firewall rules were applied to block malicious IP traffic.
This prevents unauthorized scanning and protects the network from attacks.

Verification



A terminal window on a Kali Linux system showing the execution of Nmap scans. The window title is 'root@kali: /home/kali'. The terminal output shows two Nmap scans of the IP address 192.168.1.72. The first scan at 13:23 EDT reports that the host seems down. The second scan at 13:38 EDT, using the -Pn flag to bypass ping probes, also reports that the host is down. The terminal interface includes a top bar with window controls and a taskbar at the bottom with icons for various applications.

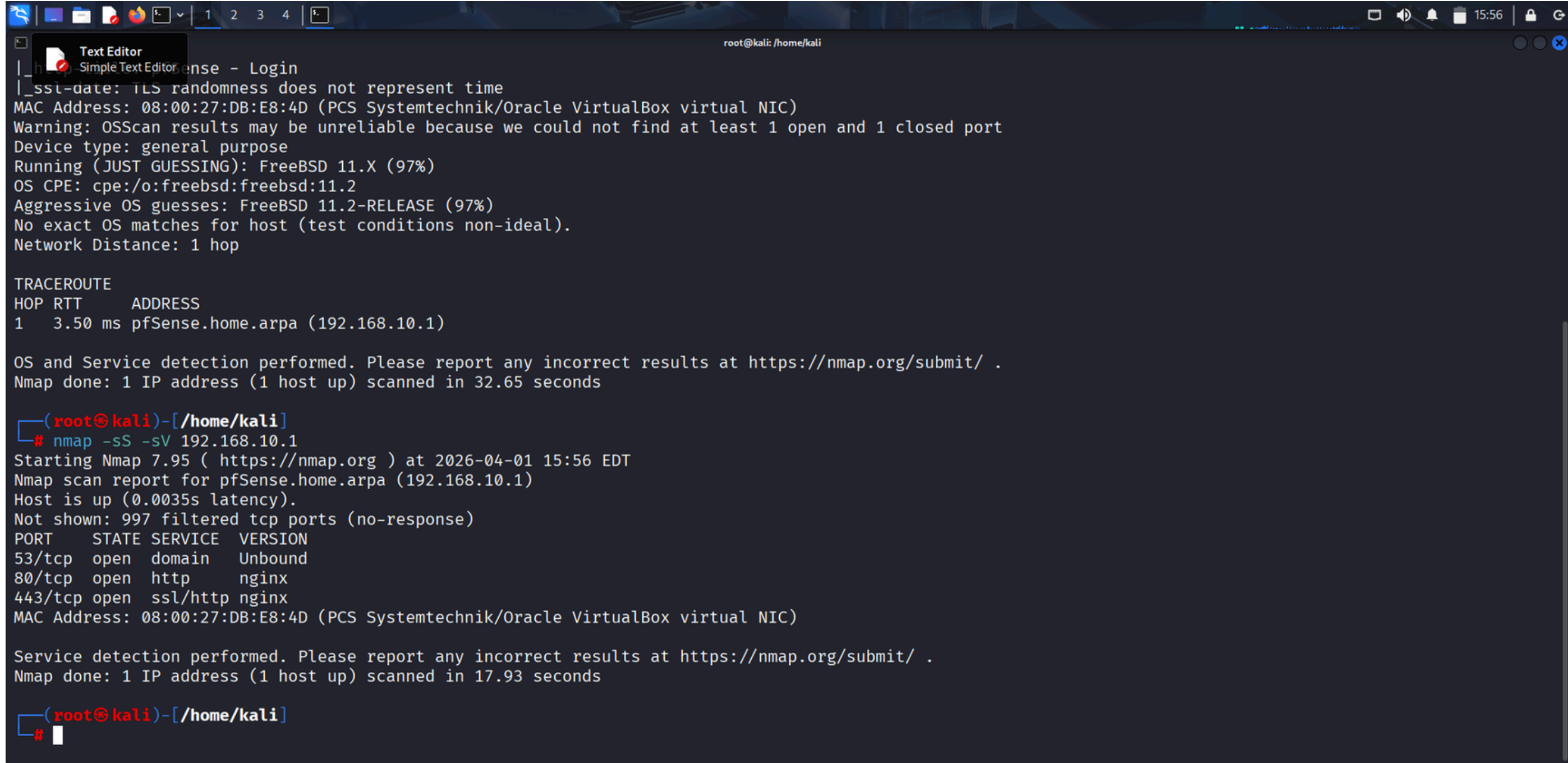
```
(root@kali)-[/home/kali]
# nmap 192.168.1.72
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-01 13:23 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.51 seconds

(root@kali)-[/home/kali]
# nmap -Pn 192.168.1.72
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-01 13:38 EDT
Nmap done: 1 IP address (0 hosts up) scanned in 1.50 seconds

(root@kali)-[/home/kali]
#
```

After applying firewall rules, the attack was re-tested.
The scan results showed blocked or filtered ports, confirming successful protection.

IDS/IPS, Suricata

A terminal window on a Kali Linux system showing the output of an Nmap scan. The terminal has a dark background with light-colored text. At the top, there's a window title bar for a 'Text Editor' and a terminal title bar showing 'root@kali: /home/kali'. The output of the scan is displayed in a light blue font. It starts with a 'Simple Text Editor - Login' message, followed by various system information like MAC address, warning about OSScan results, device type, and OS guesses. Then, it shows a 'TRACEROUTE' section with hop, RTT, and address details. This is followed by a message about OS and service detection, and the completion of the first scan. The user then runs a second Nmap command with different options, and the output shows the start of the second scan, the scan report for pfSense, and the detection of open ports (53/tcp, 80/tcp, 443/tcp) and services (Unbound, nginx). The terminal ends with the user's prompt and a cursor.

```
(root@kali)-[/home/kali]
# nmap -sS -sV 192.168.10.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-01 15:56 EDT
Nmap scan report for pfSense.home.arpa (192.168.10.1)
Host is up (0.0035s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain Unbound
80/tcp    open  http   nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:DB:E8:4D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.93 seconds

(root@kali)-[/home/kali]
#
```

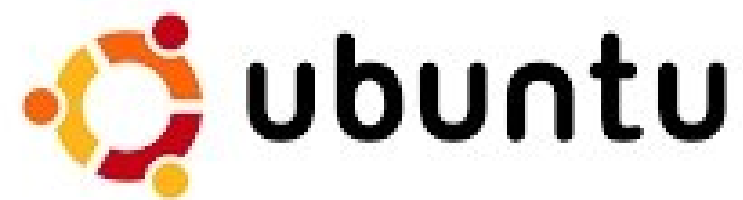
Suricata was implemented as a real-time Intrusion Detection and Prevention System (IDS/IPS) integrated with pfSense to analyze network traffic using advanced rule-based inspection. It identifies potential threats such as scanning and intrusion attempts, generates alerts, and actively blocks malicious sources. This enhances network visibility, strengthens security controls, and provides automated threat mitigation for small business environments.

Suricata monitors network traffic in real time to detect suspicious activities using security rules. It not only detects threats but also automatically blocks attackers, providing active network protection.

Tools Used



pfSense –	Firewall and network security
Kali Linux –	Attack simulation
Ubuntu –	Target system
Suricata –	Intrusion detection



CONCLUSION

This project implements a practical Cybersecurity framework for small businesses using pfSense and Suricata to detect and prevent network-based attacks. A virtual lab environment (Kali Linux, Ubuntu, pfSense) was used to simulate real-world threats such as network scanning and intrusion attempts. Suricata was configured with updated rule sets to monitor traffic, generate alerts, and automatically block malicious activity. The solution focuses on cost-effective security controls, including intrusion detection, traffic filtering, and threat mitigation, providing a scalable and efficient defense mechanism for small-scale business networks.

And also this project proves that small businesses can achieve effective cybersecurity using structured frameworks and open-source tools.

By implementing essential controls and proactive monitoring, organizations can significantly reduce risks and improve overall security posture.

Thank You !